Oscar I. Hernandez

Summary

I'm looking for an immediate job placement, hopefully one where I can apply my strong background in mathematics and computer science.

Email: oscar.hernandez@simons-rock.edu Website: https://mathemonads.github.io
Phone: +1 (646) 961-3715 GitHub: http://github.com/mathemonads

Education

Bard College at Simon's Rock (BCSR) Aug 2013 – May 2017
BA, Mathematics and Computer Science, Honors Thesis [HW17] – May 2017
AA, Liberal Arts and Sciences Aug 2013 – May 2015

Awards: W.E.B. Du Bois Scholarship (2013–2017), Alumni Scholarship (2013–2017), Supplemental Return Grant (2014–2017), Dean's List (Fall 2015 and Fall 2016).

Experience

Freelance Software Engineer	Independent Contractor	Jan 2018 –
Founder & Robotics Engineer	ARKS	Apr 2018 – Jan 2020
Research Assistant	BCSR: Economics	Spring 2016
Research Intern	BCSR: Computer Science	Summers 2015 & 2016

Teaching

5th Grade Humanities Teacher	SACS HNW	Jul 2017 – Dec 2017
G5 World History	G5 Math	G5–6 CS Quarter Elective
G5 English Language Arts	G5 Earth Science	G5–8 CS Semester Elective
Subject/Class Tutor	BCSR	Fall 2014 – Spring 2017
Computer Science	Latin I	Intro to Computer Science
Critical Reasoning	Formal Logic	Algorithms & Data Structures
Algebra I with Geometry	Discrete Math	Graph Theory & Combinatorics
Algebra II and Trigonometry	Vector Calculus	

Awards

1. Intel Edge AI and Bertelsmann Technology (AI Track) Scholarships (Udacity 2019)
2. Machine Learning with Tensorflow on GCP Specialization (Coursera 2020)
3. NSF I-Corps grant recipient x2 (UCR Spring 2019, CSUN Fall 2019)
4. Best Social/Civic Hack (Citrus Hack 2018),
   Superstar Award & Effective Use of Technology Award (Hack SRC 2017).

Publications

[Her+18] Oscar I. Hernandez et al. "Constructing de Bruijn sequences with co-lexicographic order: The k-ary Grandmama sequence". In: *European Journal of Combinatorics* 72 (2018), pp. 1–11. ISSN: 0195-6698. URL: http://www.sciencedirect.com/science/article/pii/S0195669818300696.

[Has+17] Harold M. Hastings et al. "Dynamics of Biomimetic Electronic Artificial Neural Networks". In: *Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics (ICAND 2016)*. Ed. by Visarath In, Patrick Longhini, and Antonio Palacios. Cham: Springer International Publishing, 2017, pp. 195–207. ISBN: 978-3-319-52621-8. URL: https://link.springer.com/chapter/10.1007%5C%2F978-3-319-52621-8_18.

[HW17] Oscar I Hernandez and Aaron Williams. "Constructing Maximum-Rotating de Bruijn Sequences using Necklace Trees". *Bachelor of Arts* Senior Thesis awarded Honors. Bard College at Simon's Rock: Division of Science, Mathematics, and Computing, May 2017. URL: https://digitalcommons.bard.edu/sr-theses/1120.

[DHW16] Patrick Baxter Dragon, Oscar I. Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings". In: *LATIN 2016: Theoretical Informatics*. Ed. by Evangelos Kranakis, Gonzalo Navarro, and Edgar Chávez. Berlin, Heidelberg: Springer Berlin Heidelberg, 2016, pp. 347–361. ISBN: 978-3-662-49529-2. URL: https://link.springer.com/chapter/10.1007/978-3-662-49529-2_26.

Workshop papers, invited presentations, and small articles:

[HY20] *Lower Gauge Theory*. Special Interest Group on Harry Q Bovik. Similar joke articles published annually since 2016. Association for Computational Heresy. Carnegie Mellon University, Apr. 2020. URL: http://sigbovik.org/2020/proceedings.pdf#page=91.

[HI19] *Zero Shot Sentiment Analysis on Tweets in Any Language*. Poster Session #1. Southern California Natural Language Processing Symposium. University of Southern California, Sept. 2019. URL: https://socalnlp.github.io/symp19/index.html.

[HT19] *Toward Compositional Transfer Learning*. Contributed Paper Session 1. American Mathematical Society Fall Western Sectional Meeting. UC Riverside, Nov. 2019. URL: http://www.ams.org/amsmtgs/2266_abstracts/1153-18-420.pdf.

[Her+19] *Spinning Around and Around: Turnstile Puzzles are* ***PSPACE****-Complete*. Work under revision started at the invite-only 32nd Bellairs Winter Workshop on Computational Geometry. 2019.

[Her+16] *Analog Artificial Neural Networks*. Poster Session. Awarded 2nd place and $200 https://www.aps.org/units/nyss/governance/minutes/upload/Fall16.pdf#page=4. New York State Section of the American Physical Society (NYSS-APS) Symposium on the Physics of 2D Materials. Rensselaer Polytechnic Institute, Nov. 2016. URL: https://simons-rock.edu/news/students-win-prize-physics-research.php.